

LogiVision: Real-Time Logistics Perception & Reasoning

PRE-HACKATHON SCREENING PRESENTATION • TRACK: COMPUTER VISION + APPLIED ML

Presenter: Aadithya R | **Model:** RT-DETR-Large | **Framework:** FastAPI + Docker + Cloudflare

Live Demo URL: <https://prices-debug-match-twist.trycloudflare.com>

Key Innovations: Non-COCO 5-class ontology, HTTP Range-request dataset streaming, 100% single-class pure dataset split, and a hand-written 3-tier confidence guardrail.

1. Domain Selection & Non-COCO Compliance

- **Industrial Domain:** Freight Consolidation Hubs & Warehouse Logistics.
- **5-Class Ontology:** cardboard box (0), forklift (1), freight container (2), wood pallet (3), truck (4).
- **Hard Constraint 3 Compliance:** *Wood pallet, freight container, and cardboard box* are **non-COCO classes**.
- **Why it matters:** Prevents off-the-shelf COCO model evaluation and forces end-to-end domain fine-tuning.

2. Dataset Acquisition & Mid-Project Engineering Pivot

- **Selective HTTP Range Streaming:** Built custom engine (scripts/prepare_dataset.py) to read remote zip Central Directory and extract images over HTTP Range requests in ~14.5 mins.
- **Mid-Project Pivot:** Initial run suffered from multi-class skew (>387 pallets vs 300 boxes) and augmented duplicate SHA256 hashes.
- **Course Correction:** Enforced strict single-class purity (`len(classes_in_file) == 1`), stem deduplication, and class re-mapping. Passed all 12 checks in `verify_dataset.py` with 0 hash collisions.

3. Model Architecture & Split Strategy

- **Model Architecture:** RT-DETR-L (Real-Time Detection Transformer with Efficient Hybrid Encoder).
- **Dataset Split (2,500 Images):**
 - **Train (60%):** 1,500 images (300 per class)
 - **Validation (20%):** 500 images (100 per class)
 - **Test (20% Held-Out):** 500 images (100 per class)
- **Hardware:** NVIDIA GeForce RTX 3050 6GB Laptop GPU (~45 ms inference speed).

4. Evaluation & Root-Cause Failure Analysis

Five Genuine Failure Cases Diagnosed:

1. **Extreme Occlusion (>70% Hidden Forklift)**: Cross-attention decoder query failure.
2. **Small Objects (Distant Box <15px)**: Downsampling via P3/8 stride.
3. **Class Confusion (Container vs. Truck Body)**: Geometry & corrugated metal ambiguity.
4. **Low Contrast (Unlit Pallets in Shadow)**: Near-zero contrast against concrete.
5. **Cluster Merging (Palletized Box Stacks)**: Plastic wrap smoothing individual borders.

5. Part B: Hand-Written Reasoning & Guardrails

- **No Agentic Frameworks (Hard Constraint 1):** Pure Python in `app/reasoning.py`.
- **Intent Routing:** Non-visual queries bypass detector; visual queries (COUNT, PRESENCE, SPATIAL) invoke RT-DETR.
- **Three-Tier Guardrail:** High (≥ 0.50), Medium (0.25-0.49), Low (< 0.25).
- **Honest Refusal Payload:** Outputs *'Insufficient information to answer confidently'* when confidence < 0.25 .